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ORIGINAL ARTICLE

Higher seasonal temperature enhances the occurrence of methicillin resistance of *Staphylococcus aureus* in house flies (*Musca domestica*) under hospital and environmental settings

Md. Abdus Sobur[1] · Md. Saiful Islam[1] · Zobayda Farzana Haque[1] · Ebiowei Samuel F. Orubu[2] · Antonio Toniolo[3] · Md. Abu Choudhury[4] · Md. Tanvir Rahman[1]

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Abstract

Antimicrobial resistance (AMR) emergence in commensal and pathogenic bacteria is a global health issue. House flies (*Musca domestica*) are considered as biological and mechanical vectors for pathogens causing nosocomial infections, including methicillin-resistant *Staphylococcus aureus* (MRSA). However, the prevalence of antimicrobial resistance and the role of temperature on the occurrence of *Staphylococcus aureus* and MRSA in house flies in a hospital environment have not been studied. A total of 400 house flies were collected in winter and summer from four hospital-associated areas in Mymensingh, Bangladesh. Detection of *S. aureus* and MRSA in flies was done by culturing, staining, and PCR methods targeting *nuc* and *mec* genes (*mecA* and *mecC*), respectively. Disc diffusion test was used to detect resistance phenotype against six antimicrobials. Logistic regression models were constructed to assess the effect of temperature on the frequency of antimicrobial resistance, and on the presence of the *nuc* and *mecA* genes, and location of samples in and around a hospital environment. By PCR, *S. aureus* was detected in 208 (52%) samples. High frequencies of resistance (≥ 80% of isolates) to amoxicillin, azithromycin, and oxacillin were observed by disk diffusion test. Increase in temperature had a positive effect on the occurrence of *S. aureus* and MRSA isolates as well as on their resistance to individual and multiple antimicrobials. Among the study areas, hospital premises had increased odds of having *S. aureus*. Increased temperature of summer significantly increased the occurrence of MRSA in house flies in and around the hospital environment, which might pose a human and animal health risk.

Introduction

The emergence of antimicrobial resistance (AMR) in nosocomial pathogens poses a global health risk by increasing mortality from minor injuries and preventable infections (Nathan and Cars 2014). It is estimated that by 2050, around ten million deaths per year will occur due to antibiotic-resistant pathogens (de Kraker et al. 2016). Previous studies revealed associations among increase in temperature, bacterial growth (Ratkowsky et al. 1982) and AMR development (MacFadden et al. 2018). We hypothesize that the effects of global warming, especially increase in temperature, will enhance the emergence of antibiotic-resistant pathogens (MacFadden et al. 2018). In Bangladesh, difference in temperature between summer (about 30–40 °C) and winter (about 10 °C) is significant, which might affect the emergence of AMR pathogens (Dailiana et al. 2008; Elegbe 1983; Guo et al. 2018).

Penicillin, the first antibiotic, originally had high efficacy against staphylococcal infections, but soon developed resistance (Kirby 1944; Lowy 1998), resulting in a global health problem within the next two decades (Rountree et al. 1954). Methicillin, a penicillin derivative, was introduced in 1959 to combat penicillin-resistant staphylococci (Jevons and Parker 1964). Methicillin-resistant *S. aureus* (MRSA) was first reported from a hospital in the UK (Lowy 1998; Jevons

✉ Md. Tanvir Rahman
tanvirahman@bau.edu.bd

[1] Department of Microbiology and Hygiene, Faculty of Veterinary Science, Bangladesh Agricultural University, Mymensingh 2202, Bangladesh

[2] Institute for Health System Innovation and Policy, Boston University, 180 Riverway, Boston, MA 02215, USA

[3] Global Virus Network, University of Insubria, 21100 Varese, Italy

[4] Menzies Health Institute Queensland, Griffith University, 170 Kessels Road, Nathan, Queensland 4111, Australia

1961). The mobile genetic element mec (*SCCmec*) harboring the *mecA*, *mecB*, and *mecC* genes is responsible for the development of MRSA (Matsuhashi et al. 1986). MRSA is a zoonotic pathogen that is common in hospitals, community settings, and livestock farms (Chatterjee and Otto 2013), and it is a well-known cause of nosocomial surgical wound infections. A US study has demonstrated a higher mortality attributable to MRSA infections than of AIDS (Klevens et al. 2007). Along with methicillin resistance, *S. aureus* exhibits increased resistance to several other classes of antibiotics that limits treatment options (Abdel-Moein and Zaher 2019; Kadariya et al. 2019; Hiramatsu et al. 2014). The emergence of multidrug resistance in *S. aureus* isolates poses a public health threat worldwide (Kadariya et al. 2019; Hiramatsu et al. 2014). Furthermore, multidrug-resistant (MDR) *S. aureus* has become a community-acquired infection (Kadariya et al. 2019).

House flies (*Musca domestica)* are effective mechanical and biological vectors for the transmission of zoonotic pathogens, including AMR bacteria (Jones et al. 2008; Olsen 1998; Rahuma et al., 2005). Recent studies from Bangladesh have reported that house flies collected from various environments including hospitals carried MDR *E. coli* and *Salmonella* (Sobur et al. 2019a, 2019b, 2019c). Being often part of human skin microbiota, *S. aureus* has been successfully isolated from flies (Rahuma et al., 2005; Graham et al. 2009; Nazari et al. 2017; Poudel et al. 2019; Khamesipour et al. 2018; Onwugamba et al. 2020). Flies are more abundant in places of human activities such as hospitals, food corners, restaurants, fruit shops, slaughterhouses, livestock, and poultry farms (Awache and Farouk 2016). As the diet and reproduction of house flies depend on feces, garbage, rooting fruits, spoiled meats, and other decaying organic substances, they can easily pick up pathogens through surfaces using their body hairs or tarsi and transfer them to foods. In addition, they can transfer these pathogens by vomiting and defecation where they land. Moreover, house flies living in and around human habitats harbor a variety of pathogenic and nonpathogenic agents from several taxa including antibiotic-resistant bacteria (Bunchu et al. 2014; Chaiwong et al. 2014; Cohen et al. 1991). Types of pathogens, carried by flies, are determined by the people and animals present in where flies are found, as well as the extent to which antibiotics are utilized in any given area (Nazari et al. 2017; Davari et al. 2010; Doud and Zurek 2012; Hemmatinezhad et al. 2015; Kassiri et al. 2015; Wang 2013; Zurek and Ghosh 2014). Moreover, house flies living in the hospital environment may be associated with the spread of nosocomial infections (Hemmatinezhad et al. 2015; Sarwar 2015). In addition, a similar genetic background between *S. aureus* from humans and flies was observed; however, transmission patterns are not established yet (Schaumburg et al. 2016). Several interventional studies have showed that minimizing fly density markedly reduces the incidence of diarrhea, shigellosis, and other foodborne diseases in hospitals (Cohen et al. 1991; Chavasse et al. 1999).

Bangladesh is located in the Tropic of Cancer and is directly affected by radiation and high temperatures, where the impact of high summer temperature is demonstrated through natural calamities such as drought, Nor'westers, and cyclones. In this country, the average temperature variation between winter and summer is significant (> 15 °C). In April and May, the temperature mostly lies between 30 and 36 °C and may reach up to 40 °C (Khatun and Rashid 2016), which might increase environmental bacterial load, diversity, and their AMR profiles. The abundance of house flies, which can spread AMR bacteria such as MRSA, may be affected by rising temperatures (Onwugamba et al. 2018). To elucidate this hypothesis, our study aims to investigate the influence of seasonal temperature on the occurrence and distribution of *Staphylococcus aureus* and MRSA along with their antimicrobial resistance profiles in house flies collected in and around hospital environments.

Materials and methods

Sample size determination

Since previous work was not done in Bangladesh, we could assume the prevalence of *S. aureus* as 50%. The formula we used here to calculate the sample size was as follows (Thrusfield 1995): $n = Z^2pq/d^2$, where n = desired sample size; Z = the standard normal deviation, usually set at 1.96 at 5% level, which corresponds to 95% confidence level; p = prevalence (50% or 0.5); $q = 1 - p = (1 - 0.5) = 0.5$; and d = precision (5% so $d = 0.05$). So, $n = (1.96)^2 * 0.5 * 0.5/(0.05)^2 = 384.16$. Therefore, a total 400 flies were captured from four hospital-associated areas.

Sample collection time

We examined Bangladesh meteorological websites (http://live3.bmd.gov.bd) to identify months with maximum and minimum temperatures. December–January were the coldest (winter season), while April–May were the warmest (summer season). Samples were collected in winter (December, 2017–January, 2018) and summer (April–May, 2018). We chose five common dates as 2nd, 9th, 16th, 23rd, and 30th for each month to collect samples. For uniformity, in each day, samples were collected from 10 AM to 12 PM. A thermometer was used to record the temperature at the point of sample collection.

Sample collection

A total of 400 house flies were collected from four different locations, namely hospital premises, hospital canteen, and food centers and dustbins around the hospital in Mymensingh city. Study areas were abundant with house flies. From each place, 50 flies were collected randomly at each indicated time. On each date, 20 samples were collected from the four study areas (five samples per location).

A sterile zip-locked bag was placed into each glass beaker. The glass beakers were placed three feet above the surface. Five beakers were placed in distanced places in each study area keeping the zip of bag unlocked to allow flies enter into the bags. Once a single fly has entered into the bag, the zip was locked. Then, the bag containing the fly was removed from the beaker, and the procedure was repeated. Collected flies were immediately transferred to the laboratory at the Department of Microbiology and Hygiene, Bangladesh Agricultural University for further processing.

Sample processing and isolation of *S. aureus*

Collected flies were kept at – 20° C for 1 h to be euthanized (Vazirianzadeh et al. 2008). Before processing, each fly was examined under a stereomicroscope to confirm the species as *M. domestica* (Crosskey and Lane 1993). Individual flies were placed into an Eppendorf tube containing 1 mL nutrient broth (HiMedia, India) and crushed with sterile forceps. The medium was incubated aerobically at 37 °C overnight (Parvez et al. 2016).

After incubation, a loopful of nutrient broth from each sample was streaked onto separate mannitol salt agar (MSA) plates (HiMedia, India) and incubated aerobically at 37 °C for 48 h. Isolates producing yellow colonies on MSA agar plates were subjected to sub-culture on MSA to obtain pure cultures. Isolation and identification of *S. aureus* were performed based on morphology, Gram staining, catalase and coagulase tests, and a PCR method to detect the *nuc* gene (Kalorey et al. 2007).

Genomic DNA extraction and identification of *S. aureus*

Genomic DNA for PCR was extracted by the boiling method as described previously (Dashti et al. 2009). In brief, a pure colony was put into an Eppendorf tube and suspended in 100 µL deionized water. The tube was then boiled for 10 min, followed by immediate transfer into ice for cold shock (10 min), and centrifuged at $9200 \times g$ for 10 min. Supernatant from each tube was collected and used as PCR template. The extracted DNA was stored at – 20 °C until use. Detection of *S. aureus* was confirmed by *nuc* gene PCR (Table 1).

PCR assays were run in a final volume of 25 µL with 12.5 µL master mixture 2X (Promega, USA), 2 µL of genomic DNA, 1 µL each primer, and 8.5 µL nuclease free water. PCR-positive controls were collected from the previous study (Pondit et al. 2018), and PBS instead of genomic DNA was used as negative controls. Amplified products were analyzed by electrophoresis in 1.5% agarose gel stained with ethidium bromide. Bands were visualized under ultraviolet trans-illuminator (Biometra, Germany) using a 100 bp DNA ladder (Promega, USA) as molecular weight marker.

Antimicrobial susceptibility test

Antimicrobial susceptibility tests were performed by the Kirby-Bauer method (Bauer et al. 1966). Six commonly used antibiotics of 4 different classes were used: beta-lactams (amoxicillin, 10 µg; oxacillin, 1 µg), macrolides (azithromycin, 15 µg), fluoroquinolones (ciprofloxacin, 5 µg; levofloxacin, 5 µg), and aminoglycosides (gentamycin, 10 µg) (HiMedia, India). Tests were performed on Mueller–Hinton agar (HiMedia, India) plates inoculated with a concentration of bacterial culture equivalent to 0.5 McFarland standards and incubated for 18–24 h at 37 °C aerobically. Results were interpreted according to Clinical and Laboratory Standards Institute guidelines (CLSI 2017). Isolates showing resistance against three or more antimicrobial categories were considered as MDR (Magiorakos et al. 2012).

Table 1 List of primers used in the study to detect *Staphylococcus aureus* and methicillin-resistant *S. aureus* (MRSA) from house flies

Primer name	Gene targeted	Primer sequence (5′-3′)	Amplicon size (bp)	Reference
nuc	*nuc* F	GCGATTGATGGTGATACGGT	279	Kalorey et al. 2007
	nuc R	AGCCAAGCCTTGACGAACTAAAGC		
mecA	*mecA*F	AAAATCGATGGTAAAGGTTGG	533	Lee 2003
	*mecA*R	AGTTCTGGCACTACCGGATTTGC		
mecC	*mecC*-P1	GAAAAAAAGGCTTAGAACGCCTC	138	Stegger et al. 2012
	mecC-P2	GAAGATCTTTTCCGTTTTCAGC		

Table 2 Daily temperature at the point sample collection

Sampling date	House fly collection season			
	Winter		Summer	
	Tem°C (December, 17)	Tem°C (January, 18)	Tem°C (April, 18)	Tem°C (May, 18)
2	16	13	31	30
9	16	12	33	31
16	17	13	30	31
23	18	15	33	33
30	17	14	32	31
Average	16.8	13.4	31.8	31.2
	15.1		31.5	
Difference	16.4			

Tem temperature

Molecular detection of MRSA

S. aureus isolates were further characterized by PCR targeting *mec*A and *mec*C genes to confirm the isolates as MRSA. PCR is done as described earlier using the primers in Table 1.

Statistical analysis

Descriptive statistics

Antimicrobial susceptibility data for *S. aureus*-positive isolates were inputted into an Excel spreadsheet and subsequently transferred into a statistical software program (STATA Intercooled, version 14.2, Stata Corporation, College Station, TX) for further analysis. Estimates of the proportion of *S. aureus*-positive isolates that were resistant to each of the six tested antimicrobials were calculated by dividing the number of isolates resistant to an antimicrobial by the total number of isolates tested for the antimicrobial. For all estimates, exact binomial 95% confidence intervals (CIs) were calculated.

Regression analysis

To determine the effect of temperature on the presence of antimicrobial resistance, logistic regression models were constructed. The outcome variable consisted of the presence of antimicrobial resistance, while temperature in Celsius (continuous variable) was the independent variable.

To determine the effect of temperature and sampling location on the presence of the *nuc* and *mecA* genes, separate logistic regression models were built. The outcome variable consisted of the presence/absence (binomial variable) of the *nuc* or *mecA* genes. Temperature in Celsius (continuous variable) and sampling location (categorical variable) were included as independent (i.e., predictor) variables, while the location categorical variable included the four sampling locations: dustbin, food center around hospital, hospital premises, and hospital canteen. The location category with the lowest odds was chosen as the reference to which all the other categories were compared.

To identify associations among the presence of *mecA* gene and antimicrobial resistance, individual univariable logistic regression models were built. The binary (yes/no) outcome variable represented the presence of the *mecA* gene, while the independent variable was the frequency of resistance to the antimicrobial.

Results

Seasonal temperature

Average temperature of December, 2017, and January, 2018, was respectively 16.8 °C and 13.4 °C. Thus, 15.1 °C was the mean winter temperature. In 2018, the

Table 3 Effect of temperature and sampling location on the presence of *nuc* gene in *S. aureus* isolates

Categories	Variable	Odds ratio	[95% confidence interval]	P-value
Temperature	Change in temperature	1.08	1.05–1.11	< 0.001
Areas	Food center around hospital	Reference	-	-
	Dustbin	2.70	1.48–4.92	0.001
	Hospital	4.52	2.43–8.41	< 0.001
	Hospital canteen	1.81	1.00–3.28	0.052

mean summer temperature was 31.5 °C (April 31.8 °C; May 31.2 °C). The difference in temperature between the winter and summer was 16.4 °C (Table 2).

Seasonal variation in the prevalence of *nuc* gene-confirmed *S. aureus* isolates

Among the 400 house flies analyzed, 208 (52%) were *S. aureus*-positive. Prevalence of confirmed *S. aureus* was higher in summer (64%, 128/200) compared to winter (40%, 80/200) season. With increase in temperature, the odds of detecting the *nuc* gene significantly increased (OR = 1.08). Hospital (OR = 4.52), dustbin (OR = 2.70), and hospital canteen (OR = 1.81) locations compared to the food center around the hospital (referent) had an increased odd of the *nuc* gene presence (Table 3). Among the four collection locations, flies at hospital premises were carrying higher numbers of *S. aureus* isolates.

Antibiotic resistance of *S. aureus* isolates

In the *S. aureus* isolates, there was a high frequency of resistance (≥ 80% of isolates) to amoxicillin, azithromycin, and oxacillin and a moderate frequency of resistance (23–39% of isolates) to ciprofloxacin, gentamicin, and levofloxacin (Table 4).

Temperature was positively associated with the presence of antimicrobial resistance to individual and multiple antimicrobials. With increase in temperature, the odds of antimicrobial resistance increased for amoxicillin [OR = 1.08 (CI: 1.02–1.14), $p = 0.005$], azithromycin [OR = 1.06 (CI: 1.00–1.11), $p = 0.020$], ciprofloxacin [OR = 1.09 (CI: 1.05–1.14), $p < 0.001$], levofloxacin [OR = 1.09 (CI: 1.03–1.14), $p = 0.001$], and oxacillin [OR = 1.07 (CI: 1.02–1.12), $p = 0.005$]. No association was detected between the presence of resistance to gentamicin and temperature.

Table 4 Prevalence of *S. aureus* isolates that were resistant to six selected antimicrobials

Antimicrobial class	Antimicrobial[A]	n(%)[B] [CI][C]
Aminoglycosides	GEN	48 (23.08) [17.53–29.41]
β-Lactams	AMO	184 (88.46) [83.32–92.47]
	OX	175 (84.13) [78.45–88.82]
Macrolides	AZM	176 (84.62) [78.98–89.23]
Quinolones	CIP	80 (38.46) [31.82–45.44]
	L	47 (22.60) [17.10–28.89]

[A]*AMO* amoxicillin, *AZM* azithromycin,*CIP* ciprofloxacin, *GEN* gentamicin, *L* levofloxacin, *OX* oxacillin,[B]number and percentage of isolates resistant to the antimicrobial, [C]*CI* exact binomial 95% confidenceinterval for the percentage of isolates resistant to the antimicrobial

Analysis of collection sites showed that the antibiotic-resistant *S. aureus* isolates were more prevalent in hospital premises, except for azithromycin (Fig. 1).

Detection of MDR *S. aureus*

Multidrug resistance was positively associated with increase in temperature [OR = 1.10 (CI: 1.06–1.14), $p < 0.001$]. MDR *S. aureus* isolates were more prevalent in hospital premises (Table 5). Most AMR profiles included oxacillin resistance, as MRSA is inherently resistant to this antimicrobial. The most prevalent profile was AMO-AZM-CIP-OX (21 isolates) followed by AMO-AZM-CIP-L-OX (19 isolates) and AMO-AZM-CIP-GEN-OX (8 isolates). Interestingly, all the above profiles were identified from summer isolates. In contrast, during winter, the highest prevalent AMR profile was AMO-AZM-CIP-OX (6 isolates) (Table 5).

Molecular detection of *mecA* and *mecC* determinants

Among 175 oxacillin-resistant *S. aureus* isolates, 101 (57.7%) were *mecA*-positive. No isolates were found to harbor the *mecC* gene. Genotypic MRSA was found higher in summer (62.3%, 71/114) than that of winter (49.1%, 30/61) season. *S. aureus*-carrying *mecA* were predominant in hospital premises. With increase in temperature, the odds of detecting the *mecA* gene significantly increased [OR = 1.06 (CI: 1.02–1.10), $p = 0.001$]. No association was detected between the presence of the *mecA* gene and the source of isolates.

With the presence of the *mecA* gene, the odds of resistance increased for amoxicillin [OR = 7.98 (CI: 2.30–27.67), $p = 0.001$], azithromycin [OR = 8.59 (CI: 2.89–25.54), $p < 0.01$], ciprofloxacin [OR = 4.30 (CI: 2.36–7.84), $p < 0.001$], and levofloxacin [OR = 2.84 (CI: 1.43–5.66), $p = 0.003$]; conversely, gentamicin resistance revealed the decreased odds ratio [OR = 0.50 (CI: 0.25–0.97), $p = 0.040$]. Associations between oxacillin resistance and the presence of the *mecA* gene were not tested as *mecA* is a representative gene for oxacillin-resistant isolates.

Discussion

The World Health Organization (WHO) identified climate change as a major driver of emerging infectious diseases (WHO 2017). Environmental factors such as seasonal variation in temperature are a major contributing element in the dynamics of microbial populations. In fact, almost all

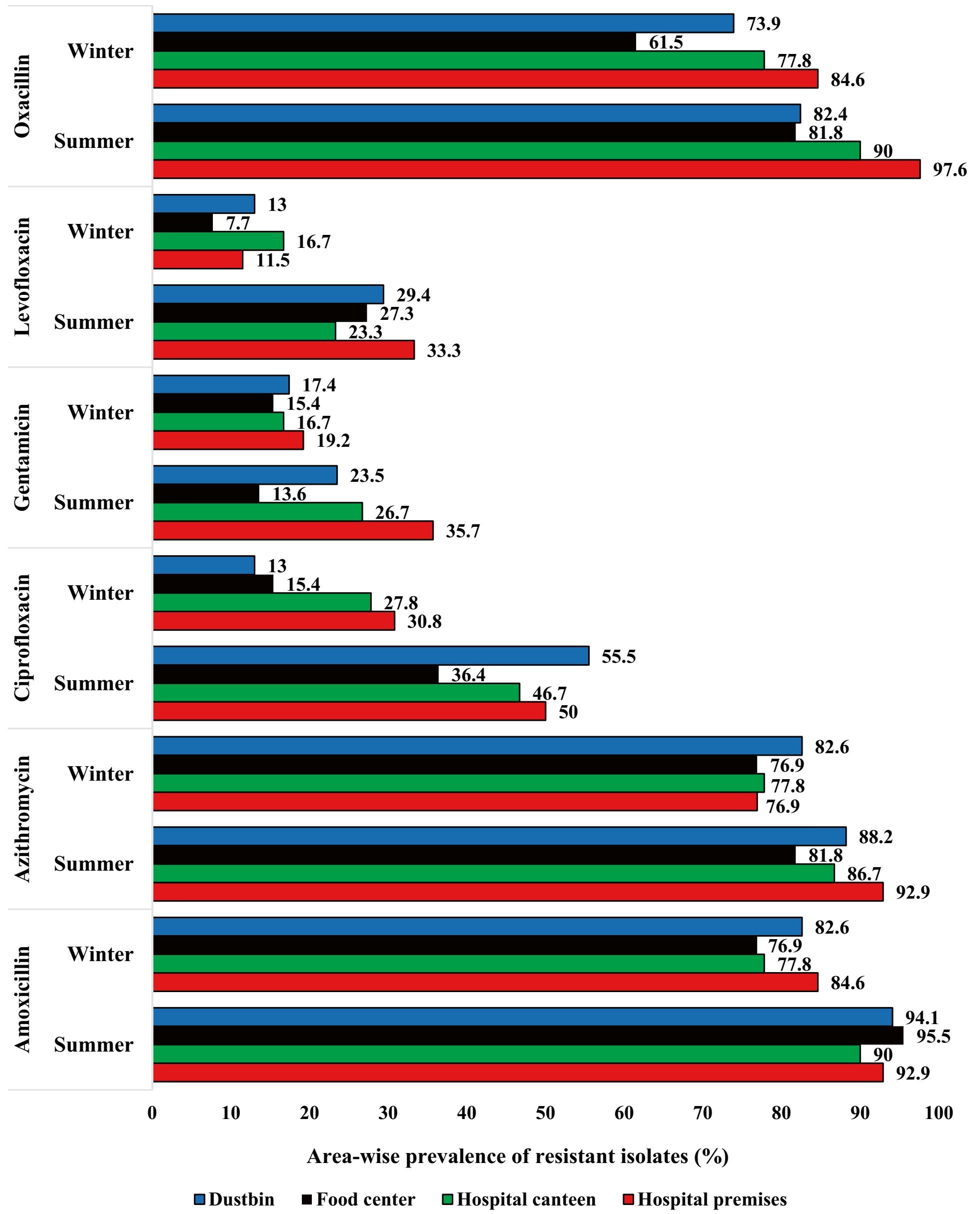

Fig. 1 Seasonal variation of resistance *S. aureus* in hospital-associated areas

the components of life on earth are under environmental influence. Global warming is a threatening issue for the earth system. Antibiotic resistance and climate change are deadly combination for all of the one-health components. Temperature increased with the result of global warming is linked with the bacterial infections and their antibiotic resistance capabilities. Augmented temperature increases horizontal gene transfer, a crucial mechanism for the acquisition of antibiotic resistance (Burnham 2021).

S. aureus is one of the most common opportunistic pathogens for humans resulting in substantial morbidity and mortality (Von Eiff et al. 2001). Our study found that 52% of house flies collected in hospital premises were carrying *S. aureus*. The intimate association between *Staphylococcus* spp. and flying insect has been observed in a recent report from the UK where 19% of insects were positive for staphylococci (Boiocchi et al. 2019). In Iran, the prevalence of *S. aureus* was estimated at only 6.3% (Nazari et al. 2017). In India, the prevalence of *S. aureus* has been reported at 36.9% in flies in hospital sites (Fotedar et al. 1991). The observed higher prevalence of *S. aureus* in this study is possibly related to the unhygienic management of biological waste, particularly wound-treated materials and other wastes where flies can easily pick up *S. aureus*. Unregulated movement of visitors may also lead to an increased load of pathogens in the hospital. Flies present in this environment acquire living microbes either by surface contamination or by feeding and then may spread pathogens to nearby areas and humans (Onwugamba et al. 2018).

Table 5 Phenotypic resistance pattern of MDR *S. aureus* isolates in summer and winter

Pattern No	Antibiotic resistance pattern	No. of antibiotics (classes)	Isolate name	No. of MDR isolates	Overall No. of MDR isolates (%)
Summer season					
1	AZM, GEN, OX	3 (3)	SHc4, SHc17, SHc37, SD19	4	77/128 (60.2%)
2	AMO, AZM, CIP	3 (3)	SHc31, SFc38	2	
3	AMO, AZM, GEN	3 (3)	SFc4, SFc12, SFc27, SD7, SD48, SD50	6	
4	AMO, AZM, L	3 (3)	SD4	1	
5	AMO, AZM, CIP, OX	4 (3)	SH11, SH14, SH22, SH28, SH43, SH44, SH47, SH49, SHc3, SHc8, SHc21, SHc25, SHc28, SHc43, SD1, SD8, SD11, SD18, SD21, SD45	21	
6	AMO, AZM, GEN, OX	4 (3)	SH36, SH38, SH41	3	
7	AMO, GEN, L, OX	4 (3)	SH50	1	
8	AMO, AZM, L, OX	4 (3)	SHc1, SFc43, SH48	3	
9	AZM, CIP, L, OX	4 (3)	SFc16	1	
10	AMO, AZM, CIP, L	4 (3)	SD31	1	
11	AMO, AZM, CIP, L, OX	5 (3)	SH8, SH30, SH32, SH39, SH40, SH45, SHc16, SHc20, SHc45, SFc8, SFc18, SFc45, SD6, SD9, SD10, SD24, SD25, SD43, SD44	19	
12	AMO, CIP, GEN, L, OX	5 (3)	SHc10	1	
13	AMO, AZM, CIP, GEN, OX	5 (4)	SH7, SH18, SH25, SH31, SH34, SH42, SHc6, SD20	8	
14	AMO, AZM, GEN, L, OX	5 (4)	SH16, SH27	2	
15	AMO, AZM, CIP, GEN, L	5 (4)	SD41	1	
16	AMO, AZM, CIP, GEN, L, OX	6 (4)	SH24, SHc18, SHc22	3	
Winter season					
1	AMO, AZM, CIP	3 (3)	WHc3, WFc16	2	24/80 (30%)
2	AMO, AZM, GEN	3 (3)	WFc7	1	
3	AMO, AZM, L	3 (3)	WFc17, WD48	2	
4	AZM, GEN, OX	3 (3)	WFc41	1	
5	AMO, AZM, CIP, OX	4 (3)	WH3, WH4, WH12, WH14, SH21, WHc18	6	
6	AMO, AZM, GEN, OX	4 (3)	WH6, WH23, WD11	3	
7	AMO, AZM, L, OX	4 (3)	WHc14	1	
8	AMO, AZM, CIP, L	4 (3)	WD16	1	
9	AMO, AZM, CIP, L, OX	5 (3)	WH16. WH19, WHc21, WD22	4	
10	AMO, AZM, CIP, GEN	4 (4)	WH42	1	
11	AMO, AZM, CIP, GEN, OX	5 (4)	WD19	1	
12	AMO, AZM, CIP, GEN, L, OX	6 (4)	WHc19	1	
Total					101/208 (48.6%)

AMO= Amoxicillin, AZM= Azithromycin, CIP= Ciprofloxacin, GEN= Gentamicin, L= Levofloxacin, OX= Oxacillin, MDR= Multidrug resistant, S= Summer, W= Winter, H= Hospital premises, Hc= Hospital canteen, Fc= Food corner, D= Dustbin

A significant increase in staphylococcal infections in warmest months even in the tropics was previously reported (Van De Griend et al. 2009; Wiersma et al. 2009). Our study revealed that seasonal temperature is positively associated with the prevalence of *S. aureus* in house flies. Temperature is known to affect bacterial growth in vitro (Ratkowsky et al. 1982). With the maximum house fly density (March) and incidence of shigellosis (April), an increase in shigellosis cases was previously recorded in Bangladesh following 1 to 2 months of peak house fly density (Farag et al. 2013). Although there are mainly three periods, namely winter, summer, and rainy season in Bangladesh, winter (December–January) and summer (April–May) were considered for the present study. However, Onwugamba et al. (2020) showed that a lower mean temperature was associated with the presence of *S. aureus* in flies from Nigeria. This may be due to several environmental factors such as higher air pressure, lower wind pressure, and higher relative humidity, which can assist in the occurrence of *S. aureus*.

AMR is a global public health challenge, and when MDR strains occur, pathogens strongly challenge the health systems (Islam et al. 2021). MRSA are serious pathogens that cause health communities, particularly hospital-acquired infections around the globe (Van De Griend et al. 2009). In our study, 57.7% (101/175) MRSA isolates harbored *mecA* gene; no isolates were positive for *mecC* gene. Previously Cikman et al. (2019) showed that 63.8% MRSA isolates harbored *mecA* gene and all the isolates were negative for *mecC* gene which is similar to our present study. Detecting *mecA* by PCR is the gold standard method in the detection of MRSA isolates; however, in our present study, both *mecA* and *mecC* were absent in 74 (42.3%) MRSA isolates which were phenotypically resistant to oxacillin. The absence of both *mecA* and *mecC* gene in these MRSA isolates may be due to the PCR method we utilized or the presence of other *mec* genes including *mecB* and *mecD* genes (Cikman et al. 2019). In addition, *mecA*-negative MRSA isolates have been occasionally reported globally (Nomura et al. 2020). The findings from our study could lead to the discovery of other intrinsic factors that compete with the *mecA* gene in the development of MRSA.

In the present study, MRSA strains were detected more frequently in hospital premises, as previously suggested by studies of MRSA strains in Bangladesh hospitals (Hasan et al. 2016; Khanam et al. 2016). In these settings, the large number of patients and visitors might contribute to the spread of AMR strains. In addition, the wide use of antibiotics in hospitals and animal farms was linked to the emergence of MRSA strains (Chambers and DeLeo 2009; Mehndiratta and Bhalla 2014), which are priority 2 pathogens according to WHO (Tacconelli et al. 2018). In our study, amoxicillin was the least effective against *S. aureus* with 88.4% resistance. Similarly, a study at Jahangirnagar University Campus (Savar, Dhaka, Bangladesh) reported that 95% of *S. aureus* isolates from house flies were amoxicillin-resistant (Parvez et al. 2016).

Selective pressure of antimicrobial use is a major factor for AMR development (Ievy et al. 2020); however, temperature might play an important role. Our current study highlighted the effect of seasonal temperature on the occurrence of MRSA and MDR *S. aureus* strains. Antibiotic resistance was clearly higher in summer than in winter, and the findings agree with previous studies (Van De Griend 2009; Wiersma et al. 2009). It has been estimated that a 10 °C increase in temperature results in 2.7% increase of *S. aureus* antibiotic resistance (MacFadden et al. 2018). High temperatures, beyond favoring bacterial growth, were also related to increased rates of gene transfer (Lorenz and Wackernagel 1994; Walsh et al. 2011). MRSA prevalence was also higher in summer than winter in this study. Common resistance profiles of MRSA isolates from different sources and flies in summer may be associated with the prevalence of a particular strain and spread to different environments. However, findings of the present study suggested other environmental factors might be involved and require further investigation.

Conclusions

The prevalence, distribution, and diversity of pathogens and vectors in the environment are under the influence of climate and especially temperature. Our study documented a significant increase of antibiotic-resistant *S. aureus* strains in flies during summer in the hospital environments. Seasonal temperature appears to be an important environmental factor for the emergence of MRSA in house flies. However, the influence of additional climate parameters such as humidity, wind, and others for the variation in occurrence and transmission of MRSA certainly merits further investigation. Investigating the influence of climate changes on the insect-mediated spread of AMR is strongly recommended.

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Declarations

Conflict of interest The authors declare no competing interests.

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